

Kernel Methods

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Objectives of this lecture

- Turn a measure of **similarity** between data points into a learning algorithm
- Prove the **representer theorem**, which reduces an infinite-dimensional problem to n unknowns
- Characterise the similarities that hide a Hilbert structure: **positive semi-definite kernels**
- Prove **Aronszajn's theorem**, the converse of that characterisation
- Derive the **support vector machine**, its dual, and its kernel version

Notation

- $\mathcal{X}$ is the set the inputs live in. No structure is assumed on it.
- $\mathcal{H}$ is a real Hilbert space, with inner product $\langle \cdot, \cdot \rangle_{\mathcal{H}}$ and norm $\| \cdot \|_{\mathcal{H}}$. A **feature map** is a map $\varphi : \mathcal{X} \rightarrow \mathcal{H}$.
- $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ denotes a kernel, $K := (k(x_i, x_j))_{i,j}$ the **Gram matrix** of the sample, and $\mathcal{S}_n^+(\mathbb{R})$ the set of symmetric positive semi-definite matrices.
- $S := ((x_1, y_1), \dots, (x_n, y_n))$ is the training sample and $\widehat{R}_S(f) = \frac{1}{n} \sum_{i=1}^n \ell(y_i, f(x_i))$ the empirical risk.
- Anything sample-dependent carries the sample: $\widehat{R}_S$, the minimizer $\hat{\theta}_S$, its coordinates $\hat{\alpha}_S$. The dependence on the regularisation parameter $\lambda > 0$ is left implicit.
- **Abbreviation.** For $\theta \in \mathcal{H}$ we write $u(\theta) := (\langle \varphi(x_1), \theta \rangle_{\mathcal{H}}, \dots, \langle \varphi(x_n), \theta \rangle_{\mathcal{H}}) \in \mathbb{R}^n$, the vector of the n predictions on the training inputs.

I. Similarities

From similarity to prediction

Idea. When we are able to measure the similarity of the data, kernels are a way of generalising linear algorithms into non-linear methods.

If (x, y) belongs to the training set and we are shown another input x' that closely resembles x , we would like to predict a y' that closely resembles y .

Kernels do this while keeping, **implicitly**, a linear structure. The whole lecture is about the word "implicitly": the linear structure will exist, we will use it, and we will never compute in it.

Similarity functions

Definition. Let $s : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$. We say that s is a similarity function if it follows the heuristic: the larger $s(x, x')$, the more x and x' resemble each other. What "resemble" means depends on the task.

Examples. For $\mathcal{X} \subset \mathbb{R}^d$,

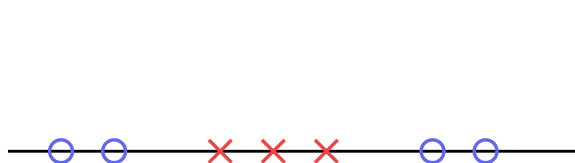
$$s(x, x') = -\|x - x'\| \quad \text{or} \quad s(x, x') = x^\top x'$$

Remark. This is a heuristic, not a mathematical definition. The next sections isolate the sub-class that is one, and that is rich enough to compute with.

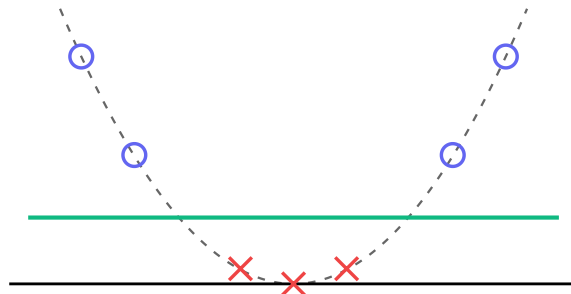
II. Motivation and the representer theorem

Raising the dimension

Idea. Many problems can be solved linearly by raising the dimension, possibly all the way to infinite dimension.



x



$\phi(x) = (x, x^2)$

On the left no threshold separates the two classes. On the right, after the feature map $\varphi(x) = (x, x^2)$, a linear rule does. The price is the dimension, and we will let it become infinite.

The model in infinite dimension

Let $\mathcal{H}$ be a Hilbert space and $\varphi : \mathcal{X} \rightarrow \mathcal{H}$ a feature map. We predict with $f_\theta(x) = \langle \varphi(x), \theta \rangle_{\mathcal{H}}$ and we minimize the regularised empirical risk over the whole of $\mathcal{H}$:

$$\hat{\theta}_S \in \arg \min_{\theta \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n \ell(y_i, \langle \varphi(x_i), \theta \rangle_{\mathcal{H}}) + \frac{\lambda}{2} \|\theta\|_{\mathcal{H}}^2$$

This is the regularised empirical risk minimization of the previous lectures, with $\Theta = \mathcal{H}$. Two obstacles: we cannot run a descent algorithm on an infinite number of coordinates, and in general we cannot even write $\varphi(x)$ down.

Representer theorem

Theorem. Let $\Psi : \mathbb{R}^n \times \mathbb{R}_+ \rightarrow \mathbb{R}$ be non-decreasing in its last argument, and let $\mathcal{H}_D := \text{span}\{\varphi(x_1), \dots, \varphi(x_n)\}$. Then

$$\inf_{\theta \in \mathcal{H}} \Psi(u(\theta), \|\theta\|_{\mathcal{H}}^2) = \inf_{\theta \in \mathcal{H}_D} \Psi(u(\theta), \|\theta\|_{\mathcal{H}}^2)$$

If the infimum is attained, it is attained at some $\hat{\theta}_S \in \mathcal{H}_D$. If Ψ is strictly increasing in its last argument, then **every** minimizer lies in $\mathcal{H}_D$.

Example. $\hat{\theta}_S \in \arg \min_{\theta \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n \ell_{\text{logistic}}(y_i \langle \varphi(x_i), \theta \rangle_{\mathcal{H}}) + \frac{\lambda}{2} \|\theta\|_{\mathcal{H}}^2$, which is of this form with $\Psi(u, t) = \frac{1}{n} \sum_i \ell_{\text{logistic}}(y_i u_i) + \frac{\lambda}{2} t$.

Proof

Write $\theta = \theta_D + \theta_D^\perp$ with $\theta_D \in \mathcal{H}_D$ and $\theta_D^\perp \in \mathcal{H}_D^\perp$. Then

$$\langle \varphi(x_i), \theta \rangle_{\mathcal{H}} = \langle \varphi(x_i), \theta_D \rangle_{\mathcal{H}} \quad \varphi(x_i) \in \mathcal{H}_D$$

$$\|\theta\|_{\mathcal{H}}^2 = \|\theta_D\|_{\mathcal{H}}^2 + \|\theta_D^\perp\|_{\mathcal{H}}^2 \quad \text{Pythagoras}$$

so that, using the monotonicity of Ψ in its last argument,

$$\Psi(u(\theta), \|\theta\|_{\mathcal{H}}^2) = \Psi(u(\theta_D), \|\theta_D\|_{\mathcal{H}}^2 + \|\theta_D^\perp\|_{\mathcal{H}}^2) \geq \Psi(u(\theta_D), \|\theta_D\|_{\mathcal{H}}^2)$$

Every $\theta \in \mathcal{H}$ is beaten by its projection $\theta_D \in \mathcal{H}_D$, hence $\inf_{\mathcal{H}} \Psi \geq \inf_{\mathcal{H}_D} \Psi$. The reverse inequality holds because $\mathcal{H}_D \subset \mathcal{H}$, so the two infima are equal.

If Ψ is strictly increasing in its last argument, the inequality above is strict as soon as $\theta_D^\perp \neq 0$, so no minimizer can have a component orthogonal to $\mathcal{H}_D$. ■

Remark. The theorem lets us look for the solution in a space of dimension at most n , and this is what makes the theoretical method implementable.

Reformulation of the problem

By the theorem we may set $\theta = \sum_{j=1}^n \alpha_j \varphi(x_j)$, a change of variables from $\theta \in \mathcal{H}$ to $\alpha \in \mathbb{R}^n$. Then

$$\langle \varphi(x_i), \theta \rangle_{\mathcal{H}} = \sum_{j=1}^n \alpha_j \langle \varphi(x_i), \varphi(x_j) \rangle_{\mathcal{H}}, \quad \|\theta\|_{\mathcal{H}}^2 = \sum_{i,j=1}^n \alpha_i \alpha_j \langle \varphi(x_i), \varphi(x_j) \rangle_{\mathcal{H}}$$

so that, with $K = (\langle \varphi(x_i), \varphi(x_j) \rangle_{\mathcal{H}})_{i,j}$,

$$\hat{\alpha}_S \in \arg \min_{\alpha \in \mathbb{R}^n} \frac{1}{n} \sum_{i=1}^n \ell(y_i, (K\alpha)_i) + \frac{\lambda}{2} \alpha^\top K \alpha$$

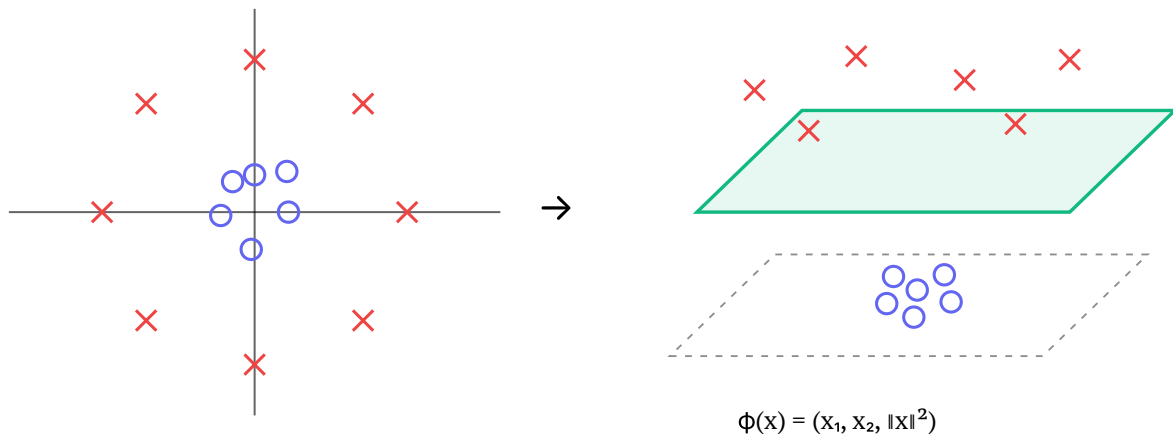
What is actually needed

To solve the problem it is enough to be able to compute the numbers

$$(\langle \varphi(x_i), \varphi(x_j) \rangle_{\mathcal{H}})_{i,j}$$

and, at prediction time, $\langle \varphi(x), \varphi(x_j) \rangle_{\mathcal{H}}$ for a new input x . The feature map itself never has to be evaluated.

This turns the question around. Instead of choosing $\mathcal{H}$ and φ , choose a function $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ directly, and ask which functions are of the form $k(x, x') = \langle \varphi(x), \varphi(x') \rangle_{\mathcal{H}}$.



III. Positive semi-definite kernels

Definition

Definition. Let $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$. We say that k is a **positive semi-definite kernel** if

$$\forall x, x' \in \mathcal{X}, \quad k(x, x') = k(x', x) \quad (\text{symmetry})$$

$$\forall n \geq 1, \forall x_1, \dots, x_n \in \mathcal{X}, \forall \alpha_1, \dots, \alpha_n \in \mathbb{R}, \quad \sum_{i,j=1}^n \alpha_i \alpha_j k(x_i, x_j) \geq 0 \quad (\text{positivity})$$

Remark. The positivity constraint says exactly that $(k(x_i, x_j))_{i,j} \in \mathcal{S}_n^+(\mathbb{R})$ for every finite family of points.

Inner products give kernels

Proposition. If $\mathcal{H}$ is a Hilbert space and $\varphi : \mathcal{X} \rightarrow \mathcal{H}$, then $k(x, x') := \langle \varphi(x), \varphi(x') \rangle_{\mathcal{H}}$ is a positive semi-definite kernel.

Proof. Symmetry is the symmetry of the inner product. For the positivity, let $n \geq 1, x_1, \dots, x_n \in \mathcal{X}$ and $\alpha_1, \dots, \alpha_n \in \mathbb{R}$:

$$\begin{aligned} \sum_{i,j=1}^n \alpha_i \alpha_j k(x_i, x_j) &= \sum_{i,j=1}^n \alpha_i \alpha_j \langle \varphi(x_i), \varphi(x_j) \rangle_{\mathcal{H}} \\ &= \left\langle \sum_{i=1}^n \alpha_i \varphi(x_i), \sum_{j=1}^n \alpha_j \varphi(x_j) \right\rangle_{\mathcal{H}} \quad \text{bilinearity} \\ &= \left\| \sum_{i=1}^n \alpha_i \varphi(x_i) \right\|_{\mathcal{H}}^2 \geq 0 \end{aligned}$$

■

Three examples

Linear kernel. $k(x, x') = x^\top x'$ on $\mathcal{X} = \mathbb{R}^d$. Positive semi-definite by the proposition, with $\varphi = \text{id}$.

Polynomial kernel. $k(x, x') = (x^\top x')^q$ for $q \geq 1$ an integer.

Gaussian kernel. $k(x, x') = \exp(-a\|x - x'\|_2^2)$ for $a > 0$.

The last two are not inner products of any obvious feature map. We need construction rules, and a theorem for the translation-invariant case.

Bochner's theorem

Theorem (Bochner). Let $q : \mathbb{R}^d \rightarrow \mathbb{R}$ be continuous. The kernel $k(x, x') := q(x - x')$ is positive semi-definite if and only if q is the inverse Fourier transform of a finite positive Borel measure ρ :

$$q(z) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} e^{iz^\top \omega} d\rho(\omega)$$

We prove the implication that is used in practice, from the integral representation to positive semi-definiteness. The converse is admitted.

Bochner's theorem, proof of the sufficient condition

Let $n \geq 1, x_1, \dots, x_n \in \mathbb{R}^d$ and $\alpha_1, \dots, \alpha_n \in \mathbb{R}$.

$$\begin{aligned}\sum_{i,j} \alpha_i \alpha_j k(x_i, x_j) &= \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \sum_{i,j} \alpha_i \alpha_j e^{i(x_i - x_j)^\top \omega} d\rho(\omega) && \text{Fubini, } \rho \text{ finite} \\ &= \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \sum_{i,j} (\alpha_i e^{ix_i^\top \omega}) \overline{(\alpha_j e^{ix_j^\top \omega})} d\rho(\omega) && \alpha_j \in \mathbb{R} \\ &= \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \left| \sum_i \alpha_i e^{ix_i^\top \omega} \right|^2 d\rho(\omega) \geq 0 && \rho \geq 0\end{aligned}$$

Symmetry holds as soon as q is even, which is the case when ρ is invariant under $\omega \mapsto -\omega$, and that is also what makes q real valued. ■

The Gaussian kernel is positive semi-definite

Apply Bochner with the Gaussian density. For every $a > 0$ and $z \in \mathbb{R}^d$,

$$e^{-a\|z\|_2^2} = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} e^{iz^\top \omega} \underbrace{(\pi/a)^{d/2} e^{-\|\omega\|_2^2/(4a)}}_{\text{density of } \rho} d\omega$$

The measure ρ is positive and finite, and invariant under $\omega \mapsto -\omega$, so $k(x, x') = \exp(-a\|x - x'\|_2^2)$ is a positive semi-definite kernel.

Construction of positive semi-definite kernels

We have seen that positive semi-definite kernels are exactly the inner products of Hilbert spaces. Here are the rules that build new ones from old.

Theorem. The set of positive semi-definite kernels on $\mathcal{X}$ is stable under

- multiplication by a non-negative real number,
- finite sums,
- products,
- pointwise limits, when the limit exists.

The first two are immediate from the definition, since a non-negative combination of non-negative quantities is non-negative, and the fourth follows by passing to the limit in the inequality $\sum_{i,j} \alpha_i \alpha_j k_m(x_i, x_j) \geq 0$.

The product is the only real argument.

Proof for the product

Let k_1, k_2 be positive semi-definite, $n \geq 1$ and $x_1, \dots, x_n \in \mathcal{X}$. Write $K_1 = (k_1(x_i, x_j))_{i,j}$ and $K_2 = (k_2(x_i, x_j))_{i,j}$. Both are in $\mathcal{S}_n^+(\mathbb{R})$, so $K_1 = U_1^\top U_1$ and $K_2 = U_2^\top U_2$. Let u_i and v_i be the i -th columns of U_1 and U_2 .

$$\begin{aligned} k_1(x_i, x_j)k_2(x_i, x_j) &= (u_i^\top u_j)(v_i^\top v_j) \\ &= \text{tr}(v_i u_i^\top u_j v_j^\top) \quad (u_i^\top u_j) \text{ scalar, } \text{tr}(v_i v_j^\top) = v_j^\top v_i \\ &= \langle u_i v_i^\top, u_j v_j^\top \rangle_F \quad \langle A, B \rangle_F = \text{tr}(A^\top B) \end{aligned}$$

The matrix $(k_1(x_i, x_j)k_2(x_i, x_j))_{i,j}$ is therefore the Gram matrix of the family $(u_i v_i^\top)_i$ for the Frobenius inner product, hence in $\mathcal{S}_n^+(\mathbb{R})$. ■

Exercises

1. Show that for every integer $q \geq 1$, the map $x, x' \mapsto (x^\top x')^q$ is a positive semi-definite kernel.
2. Show that if k is positive semi-definite, then so is $\exp(k)$.

Hints. For the first, the linear kernel is positive semi-definite and products are stable. For the second, write $\exp(k) = \lim_{m \rightarrow \infty} \sum_{p=0}^m \frac{k^p}{p!}$ and use the three other stability rules, the coefficients $1/p!$ being non-negative.

IV. Aronszajn's theorem

Statement

Theorem (Aronszajn). If $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is a positive semi-definite kernel, then there exist a Hilbert space $\mathcal{H}_k$ and a map $\varphi_k : \mathcal{X} \rightarrow \mathcal{H}_k$ such that

$$\forall x, x' \in \mathcal{X}, \quad k(x, x') = \langle \varphi_k(x), \varphi_k(x') \rangle_{\mathcal{H}_k}$$

With the proposition of the previous section, this is an equivalence: **positive semi-definite kernels are exactly the inner products of feature maps**. Choosing a kernel and choosing a pair $(\mathcal{H}, \varphi)$ are the same act, and only the kernel has to be computed.

The proof occupies the next six slides. It follows Jean-Philippe Vert.

Proof, 1) a pre-Hilbert space

For $x \in \mathcal{X}$ write $k_x := k(x, \cdot) : \mathcal{X} \rightarrow \mathbb{R}$, and consider the space of functions $\mathcal{H}_0 := \text{span}\{k_x; x \in \mathcal{X}\}$, equipped with

$$\left\langle \sum_{i=1}^m a_i k_{x_i}, \sum_{j=1}^{m'} b_j k_{z_j} \right\rangle_{\mathcal{H}_0} := \sum_{i=1}^m \sum_{j=1}^{m'} a_i b_j k(x_i, z_j)$$

This is well defined: the right-hand side equals $\sum_j b_j f(z_j)$ where $f = \sum_i a_i k_{x_i}$, so it does not depend on the way f is written, and by symmetry it does not depend on the way $g = \sum_j b_j k_{z_j}$ is written either. It is symmetric, bilinear, and positive because k is positive semi-definite.

Proof, 1) the form is definite

Reproducing property. By definition of the bracket on the generators,

$$\forall f \in \mathcal{H}_0, \forall x \in \mathcal{X}, \quad f(x) = \langle f, k_x \rangle_{\mathcal{H}_0} \quad (1)$$

Cauchy-Schwarz holds for any symmetric positive bilinear form, definiteness is not needed, so

$$\forall f \in \mathcal{H}_0, \forall x \in \mathcal{X}, \quad |f(x)| \leq \sqrt{\langle f, f \rangle_{\mathcal{H}_0}} \sqrt{k(x, x)} \quad (2)$$

Hence $\|f\|_{\mathcal{H}_0} = 0$ forces $f(x) = 0$ for every x , that is $f = 0$. So $(\mathcal{H}_0, \langle \cdot, \cdot \rangle_{\mathcal{H}_0})$ is pre-Hilbert.

Proof, 2) completion, the space

Let $(f_m)_m$ be a Cauchy sequence in $\mathcal{H}_0$. Applying (2) to $f_m - f_{m'}$ shows that for every $x \in \mathcal{X}$ the real sequence $(f_m(x))_m$ is Cauchy, hence converges.

$$\mathcal{H} := \{\text{pointwise limits of Cauchy sequences of } \mathcal{H}_0\}$$

This is a vector space of functions on $\mathcal{X}$, and it contains $\mathcal{H}_0$ through constant sequences. It remains to equip it with a Hilbert structure extending $\langle \cdot, \cdot \rangle_{\mathcal{H}_0}$.

Proof, 2) completion, the inner product

Let (f_m) and (g_m) be Cauchy in $\mathcal{H}_0$. By Cauchy-Schwarz,

$$|\langle f_m, g_m \rangle_{\mathcal{H}_0} - \langle f_{m'}, g_{m'} \rangle_{\mathcal{H}_0}| \leq \underbrace{\|f_m\|_{\mathcal{H}_0}}_{\text{bounded}} \underbrace{\|g_m - g_{m'}\|_{\mathcal{H}_0}}_{\rightarrow 0} + \underbrace{\|f_m - f_{m'}\|_{\mathcal{H}_0}}_{\rightarrow 0} \underbrace{\|g_{m'}\|_{\mathcal{H}_0}}_{\text{bounded}}$$

Cauchy sequences are bounded, so the right-hand side tends to 0 as $\min(m, m') \rightarrow \infty$. The sequence $(\langle f_m, g_m \rangle_{\mathcal{H}_0})_m$ is Cauchy in $\mathbb{R}$, hence converges, and we set $\langle f, g \rangle_{\mathcal{H}} := \lim_m \langle f_m, g_m \rangle_{\mathcal{H}_0}$ for the pointwise limits f and g . It remains to check that this does not depend on the chosen sequences.

Proof, 2) a lemma

Lemma. If (f_m) is Cauchy in $\mathcal{H}_0$ and converges pointwise to 0, then $\|f_m\|_{\mathcal{H}_0} \rightarrow 0$.

Proof. Let $\epsilon > 0$, let B bound $\|f_m\|_{\mathcal{H}_0}$, and let N be such that $\|f_m - f_N\|_{\mathcal{H}_0} \leq \epsilon$ for $m \geq N$. Writing $f_N = \sum_{i=1}^p a_i k_{x_i}$,

$$\|f_m\|_{\mathcal{H}_0}^2 = \langle f_m - f_N, f_m \rangle_{\mathcal{H}_0} + \langle f_N, f_m \rangle_{\mathcal{H}_0} \leq \epsilon B + \sum_{i=1}^p a_i f_m(x_i)$$

using Cauchy-Schwarz and then (1). The sum tends to 0 by pointwise convergence, so $\limsup_m \|f_m\|_{\mathcal{H}_0}^2 \leq \epsilon B$ for every ϵ . ■

Proof, 2) the inner product is well defined

Let $(f_m), (f'_m)$ be Cauchy in $\mathcal{H}_0$ with the same pointwise limit f , and $(g_m), (g'_m)$ with the same pointwise limit g . Then

$$|\langle f_m, g_m \rangle_{\mathcal{H}_0} - \langle f'_m, g'_m \rangle_{\mathcal{H}_0}| \leq \|f_m\|_{\mathcal{H}_0} \|g_m - g'_m\|_{\mathcal{H}_0} + \|g'_m\|_{\mathcal{H}_0} \|f_m - f'_m\|_{\mathcal{H}_0} \xrightarrow{m \rightarrow \infty} 0$$

since $(g_m - g'_m)$ and $(f_m - f'_m)$ are Cauchy and converge pointwise to 0, so the lemma applies, while the other two factors are bounded.

So $\langle \cdot, \cdot \rangle_{\mathcal{H}}$ is well defined. It is symmetric, bilinear and positive by passing to the limit. It is definite: if $\|f\|_{\mathcal{H}} = 0$ then, by (2), $|f(x)| = \lim_m |f_m(x)| \leq \lim_m \|f_m\|_{\mathcal{H}_0} \sqrt{k(x, x)} = 0$ for every x .

Proof, 2) density of $\mathcal{H}_0$

Let $f \in \mathcal{H}$ and let (f_m) be Cauchy in $\mathcal{H}_0$ converging pointwise to f . For fixed m , the sequence $(f_p - f_m)_p$ is Cauchy in $\mathcal{H}_0$ and converges pointwise to $f - f_m$, so by definition of the inner product

$$\lim_{m \rightarrow \infty} \|f - f_m\|_{\mathcal{H}} = \lim_{m \rightarrow \infty} \lim_{p \rightarrow \infty} \|f_p - f_m\|_{\mathcal{H}_0} = 0$$

the last equality because (f_m) is Cauchy in $\mathcal{H}_0$.

Hence $\mathcal{H}_0$ is dense in $\mathcal{H}$, and the norm of $\mathcal{H}$ restricted to $\mathcal{H}_0$ is the norm of $\mathcal{H}_0$.

Proof, 2) completeness

Let (f_m) be Cauchy in $\mathcal{H}$. By density, pick $f'_m \in \mathcal{H}_0$ with $\|f_m - f'_m\|_{\mathcal{H}} \rightarrow 0$. Then (f'_m) is Cauchy in $\mathcal{H}_0$:

$$\|f'_m - f'_{m'}\|_{\mathcal{H}_0} = \|f'_m - f'_{m'}\|_{\mathcal{H}} \leq \|f'_m - f_m\|_{\mathcal{H}} + \|f_m - f_{m'}\|_{\mathcal{H}} + \|f_{m'} - f'_{m'}\|_{\mathcal{H}} \leq \epsilon$$

for m, m' large enough.

So (f'_m) converges pointwise to some $f \in \mathcal{H}$, and the previous slide gives $\|f - f'_m\|_{\mathcal{H}} \rightarrow 0$. Therefore $\|f - f_m\|_{\mathcal{H}} \leq \|f - f'_m\|_{\mathcal{H}} + \|f'_m - f_m\|_{\mathcal{H}} \rightarrow 0$, and $\mathcal{H}$ is complete. ■

Proof, 3) characteristic properties

- $\forall x \in \mathcal{X}, k_x \in \mathcal{H}.$
- $\forall x \in \mathcal{X}, \forall f \in \mathcal{H}, f(x) = \langle f, k_x \rangle_{\mathcal{H}}.$ Indeed if $(f_m) \subset \mathcal{H}_0$ is Cauchy and tends pointwise to f , then
$$f(x) = \lim_m f_m(x) = \lim_m \langle f_m, k_x \rangle_{\mathcal{H}_0} = \langle f, k_x \rangle_{\mathcal{H}}.$$
- $\forall x, x' \in \mathcal{X}, k(x, x') = \langle k_x, k_{x'} \rangle_{\mathcal{H}}.$

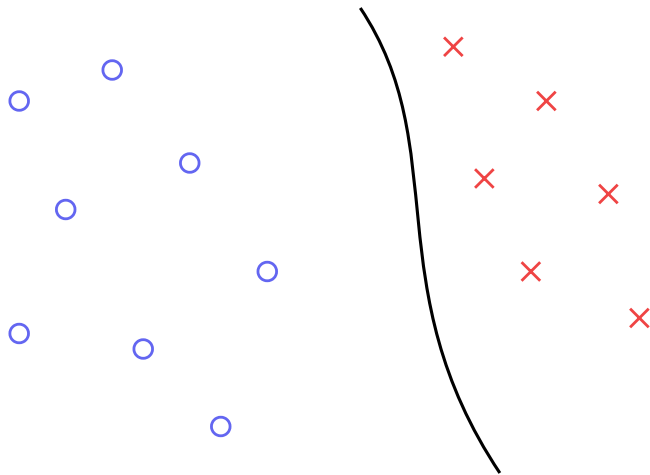
The last line is Aronszajn's theorem, with $\mathcal{H}_k = \mathcal{H}$ and $\varphi_k : x \mapsto k_x.$ ■

$\mathcal{H}$ is called the **reproducing kernel Hilbert space** of k , after the second property. Its elements are functions on $\mathcal{X}$, and evaluation at x is an inner product against k_x .

V. Support vector machines

The objective

Separate two classes, with a boundary that need not be linear.



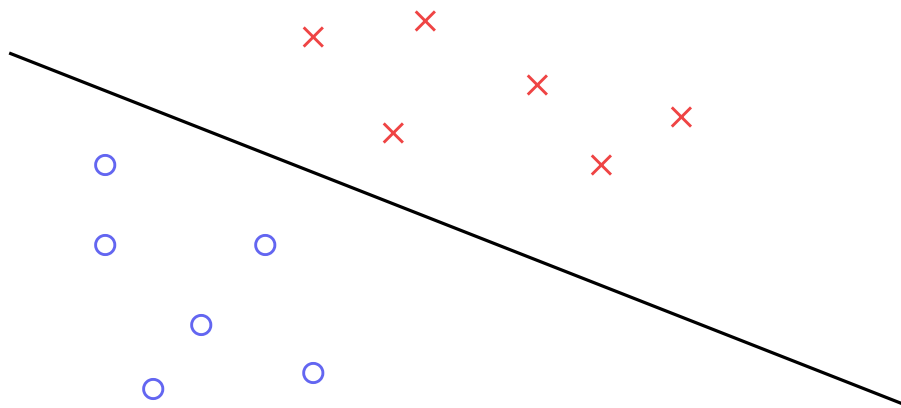
$$\hat{\theta}_S \in \arg \min_{\theta \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n \max(1 - y_i \langle \theta, \varphi(x_i) \rangle_{\mathcal{H}}, 0) + \frac{\lambda}{2} \|\theta\|_{\mathcal{H}}^2$$

Empirical risk minimization for the **hinge loss**, plus regularisation. The rest of the part builds this problem from geometry, in $\mathcal{H} = \mathbb{R}^d$ with $\varphi = \text{id}$.

1) The linearly separable case

Context. n observations $(x_i, y_i) \in \mathbb{R}^d \times \{-1, 1\}$.

Hypothesis. There exists $\theta \in \mathbb{R}^d$ such that $\forall i, y_i(\theta^\top x_i) > 0$, that is, a separating hyperplane exists.



Objective. Maximise the distance from the hyperplane to the closest point.

Distance to a hyperplane

Let $H(\theta)$ be the hyperplane of equation $\theta^\top x = 0$, for $\theta \neq 0$. Then

$$\forall x \in \mathbb{R}^d, \quad \text{dist}(x, H(\theta)) = \frac{|\theta^\top x|}{\|\theta\|_2}$$

Proof. $H(\theta)^\perp = \mathbb{R}\theta$, of which $\theta/\|\theta\|_2$ is an orthonormal basis, so

$$\text{dist}(x, H(\theta))^2 = \|x - P_{H(\theta)}(x)\|_2^2 = \|P_{H(\theta)^\perp}(x)\|_2^2 = (\theta^\top x / \|\theta\|_2)^2$$



The largest possible margin

Under the hypothesis, $y_i(\theta^\top x_i) = |\theta^\top x_i|$ for every i , so the distance from the closest point to the hyperplane is $\min_i y_i(\theta^\top x_i)/\|\theta\|_2$, and we solve

$$\max_{\theta \neq 0} \min_{i \in \{1, \dots, n\}} \frac{y_i(\theta^\top x_i)}{\|\theta\|_2}$$

An equivalent formulation

The objective is invariant under $\theta \mapsto c\theta$ for $c > 0$, and under the hypothesis its value is positive, so we may impose the normalisation $\min_i y_i(\theta^\top x_i) = 1$. Maximising $1/\|\theta\|_2$ under that constraint is minimizing $\frac{1}{2}\|\theta\|_2^2$, and the constraint can be relaxed to an inequality since it is saturated at the optimum:

$$\min_{\theta} \frac{1}{2}\|\theta\|_2^2 \quad \text{s.t.} \quad \forall i, y_i(\theta^\top x_i) \geq 1$$

A convex quadratic problem with linear constraints, in place of a maximin.

2) The linearly non-separable case

If the problem is not linearly separable, that is if for every θ there is an i with $y_i(\theta^\top x_i) \leq 0$, the constraints above cannot all hold. Introduce variables measuring the sub-optimality of each constraint:

$$\min_{\theta, \xi} \frac{1}{2} \|\theta\|_2^2 + C \sum_{i=1}^n \xi_i \quad \text{s.t.} \quad \forall i, y_i(\theta^\top x_i) \geq 1 - \xi_i, \quad \forall i, \xi_i \geq 0$$

$C > 0$ sets the price of violating a constraint. The problem is feasible for every θ , since ξ_i can always be taken large enough.

Back to the hinge loss

At fixed θ the objective is increasing in each ξ_i , so at the optimum $\xi_i = (1 - y_i(\theta^\top x_i))_+$, the smallest value the two constraints allow. Substituting, dividing by nC and setting $\lambda := 1/(nC)$:

$$\min_{\theta} \frac{1}{n} \sum_{i=1}^n (1 - y_i(\theta^\top x_i))_+ + \frac{\lambda}{2} \|\theta\|_2^2$$

The geometric problem and the regularised empirical risk minimization of the first slide of this part are the same problem, and C large means λ small, that is, little regularisation.

3) The dual problem

$$\textbf{primal:} \quad \min_{\theta, \xi} \frac{1}{2} \|\theta\|_2^2 + C \sum_{i=1}^n \xi_i \quad \text{s.t.} \quad \forall i, y_i \theta^\top x_i + \xi_i - 1 \geq 0, \quad \forall i, \xi_i \geq 0$$

a convex problem with linear constraints.

Lagrangian, with multipliers $\alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_n \geq 0$:

$$\mathcal{L}(\theta, \xi, \alpha, \beta) = \frac{1}{2} \|\theta\|_2^2 + C \sum_{i=1}^n \xi_i - \sum_{i=1}^n \alpha_i (y_i \theta^\top x_i + \xi_i - 1) - \sum_{i=1}^n \beta_i \xi_i$$

Minimizing the Lagrangian

In ξ . $\mathcal{L}$ is affine in each ξ_i , with coefficient $C - \alpha_i - \beta_i$. The infimum over $\xi \in \mathbb{R}^n$ is $-\infty$ unless $\alpha_i + \beta_i = C$ for every i , and in that case all the terms in ξ disappear.

In θ . $\nabla_{\theta} \mathcal{L} = \theta - \sum_i \alpha_i y_i x_i$, so $\theta^* = \sum_{i=1}^n \alpha_i y_i x_i$, and

$$\frac{1}{2} \|\theta^*\|_2^2 - \sum_i \alpha_i y_i \theta^{*\top} x_i + \sum_i \alpha_i = \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j=1}^n \alpha_i \alpha_j y_i y_j x_i^\top x_j$$

The dual problem

Eliminating $\beta \geq 0$ through $\alpha_i + \beta_i = C$, which is exactly $\alpha_i \leq C$:

$$\max_{\alpha} \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j=1}^n \alpha_i \alpha_j y_i y_j x_i^\top x_j \quad \text{s.t.} \quad \forall i, 0 \leq \alpha_i \leq C$$

Duality. Weak duality gives $\text{Dual} \leq \text{Primal}$. The primal has strictly feasible points, for instance $\theta = 0$ and $\xi_i = 2$, so Slater's condition applies and there is strong duality: $\text{Primal} = \text{Dual}$. The dual is a quadratic program with box constraints.

Note where the data sit: only through the inner products $x_i^\top x_j$. This is the whole opening for part VI.

VI. The kernel version

Setting

Let k be a positive semi-definite kernel, $\mathcal{H}$ the Hilbert space given by Aronszajn's theorem and φ the associated feature map. We look for

$$\hat{\theta}_S \in \arg \min_{\theta \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n (1 - y_i \langle \theta, \varphi(x_i) \rangle_{\mathcal{H}})_+ + \frac{\lambda}{2} \|\theta\|_{\mathcal{H}}^2$$

Warning. This is now an infinite-dimensional problem. It can no longer be solved by stochastic gradient descent on θ .

The representer theorem applies with $\Psi(u, t) = \frac{1}{n} \sum_i (1 - y_i u_i)_+ + \frac{\lambda}{2} t$, which is strictly increasing in t since $\lambda > 0$. So $\hat{\theta}_S \in \text{span}\{\varphi(x_1), \dots, \varphi(x_n)\}$.

Rewriting the problem

Writing $\theta = \sum_j \alpha_j \varphi(x_j)$ and $K = (k(x_i, x_j))_{i,j}$, the problem becomes finite dimensional:

$$\hat{\alpha}_S \in \arg \min_{\alpha \in \mathbb{R}^n} \frac{1}{n} \sum_{i=1}^n (1 - y_i (K\alpha)_i)_+ + \frac{\lambda}{2} \alpha^\top K \alpha$$

and the classifier is

$$x \longmapsto \operatorname{sgn} \left(\sum_{j=1}^n \hat{\alpha}_{S,j} k(x, x_j) \right)$$

Only evaluations of k are needed, at training and at prediction time.

Slack variables and Lagrangian

$$\min_{\alpha, \xi} \frac{1}{n} \sum_{i=1}^n \xi_i + \frac{\lambda}{2} \alpha^\top K \alpha \quad \text{s.t.} \quad \forall i, \xi_i \geq 1 - y_i(K\alpha)_i, \quad \forall i, \xi_i \geq 0$$

With multipliers $\mu, \nu \geq 0$, and using $\sum_i \mu_i y_i (K\alpha)_i = (\text{diag}(y)\mu)^\top K\alpha$,

$$L = \frac{1}{n} \mathbf{1}^\top \xi + \frac{\lambda}{2} \alpha^\top K \alpha - (\text{diag}(y)\mu)^\top K \alpha - (\mu + \nu)^\top \xi + \mu^\top \mathbf{1}$$

Minimizing the Lagrangian

In α . $\nabla_{\alpha} L = \lambda K \alpha - K \text{diag}(y) \mu = 0$, so

$$\alpha = \frac{\text{diag}(y) \mu}{\lambda} + \epsilon, \quad \epsilon \in \text{Ker}(K) \quad (1)$$

Such an ϵ changes neither the objective nor the constraints, since both involve α only through $K \alpha$, so we take $\epsilon = 0$.

In ξ . L is affine in ξ with coefficient $\frac{1}{n} \mathbf{1} - \mu - \nu$. The infimum is $-\infty$ unless $\frac{1}{n} \mathbf{1} - \mu - \nu = 0$, and otherwise all the terms in ξ disappear.

The dual problem

Substituting (1) with $\epsilon = 0$ and eliminating $\nu \geq 0$ through $\mu + \nu = \frac{1}{n}\mathbf{1}$:

$$\max_{\mu} \sum_{i=1}^n \mu_i - \frac{1}{2\lambda} \sum_{i,j=1}^n y_i y_j \mu_i \mu_j k(x_i, x_j) \quad \text{s.t.} \quad \forall i, 0 \leq \mu_i \leq \frac{1}{n}$$

A quadratic program with box constraints, in which the data appear only through the Gram matrix. Strong duality holds for the same reason as before: the primal has strictly feasible points, so Slater's condition applies.

Reformulation with the primal variables

By (1) with $\epsilon = 0$ we have $\mu = \lambda \text{diag}(y)\alpha$, and since $y_i^2 = 1$,

$$\sum_{i=1}^n \mu_i = \lambda y^\top \alpha, \quad \frac{1}{2\lambda} \sum_{i,j=1}^n y_i y_j \mu_i \mu_j k(x_i, x_j) = \frac{\lambda}{2} \alpha^\top K \alpha$$

Dividing the objective by $\lambda/2 > 0$, the dual problem reads

$$\max_{\alpha} 2\alpha^\top y - \alpha^\top K \alpha \quad \text{s.t.} \quad \forall i, 0 \leq y_i \alpha_i \leq \frac{1}{\lambda n}$$

The numerical resolution of this program is the subject of the lab.

What to remember

- A positive semi-definite kernel is exactly an inner product of features: one direction is a two-line computation, the other is Aronszajn's theorem
- The representer theorem turns a regularised problem posed in $\mathcal{H}$ into a problem over $\alpha \in \mathbb{R}^n$
- Everything is then computed from the Gram matrix K , and φ never appears, at training or at prediction time
- New kernels from old: non-negative combinations, products, pointwise limits, and Bochner for the translation-invariant ones
- Maximal margin, slack variables, and hinge loss plus ridge penalty are three readings of the same problem
- The kernel SVM dual is a box-constrained quadratic program where the data enter only through k